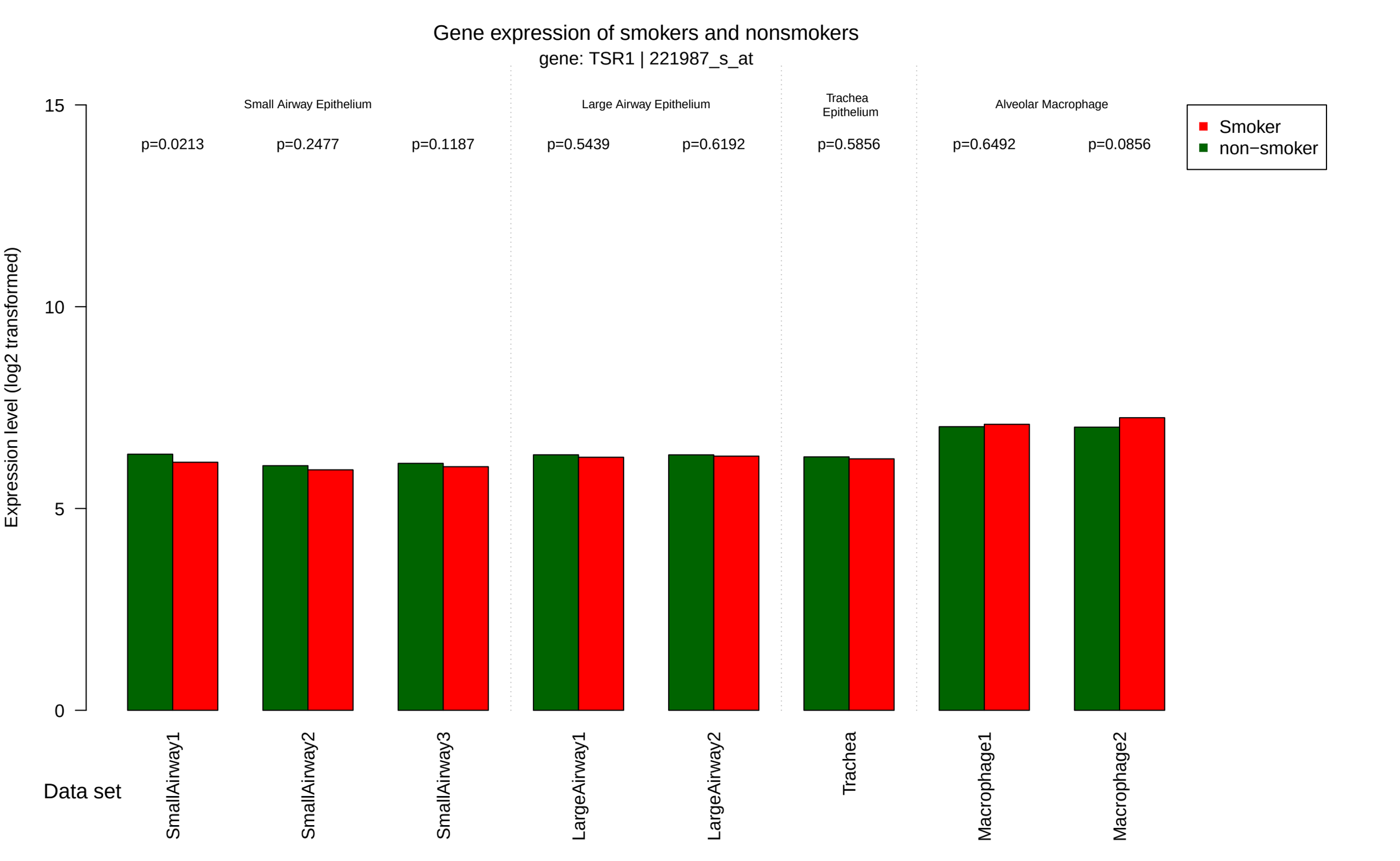
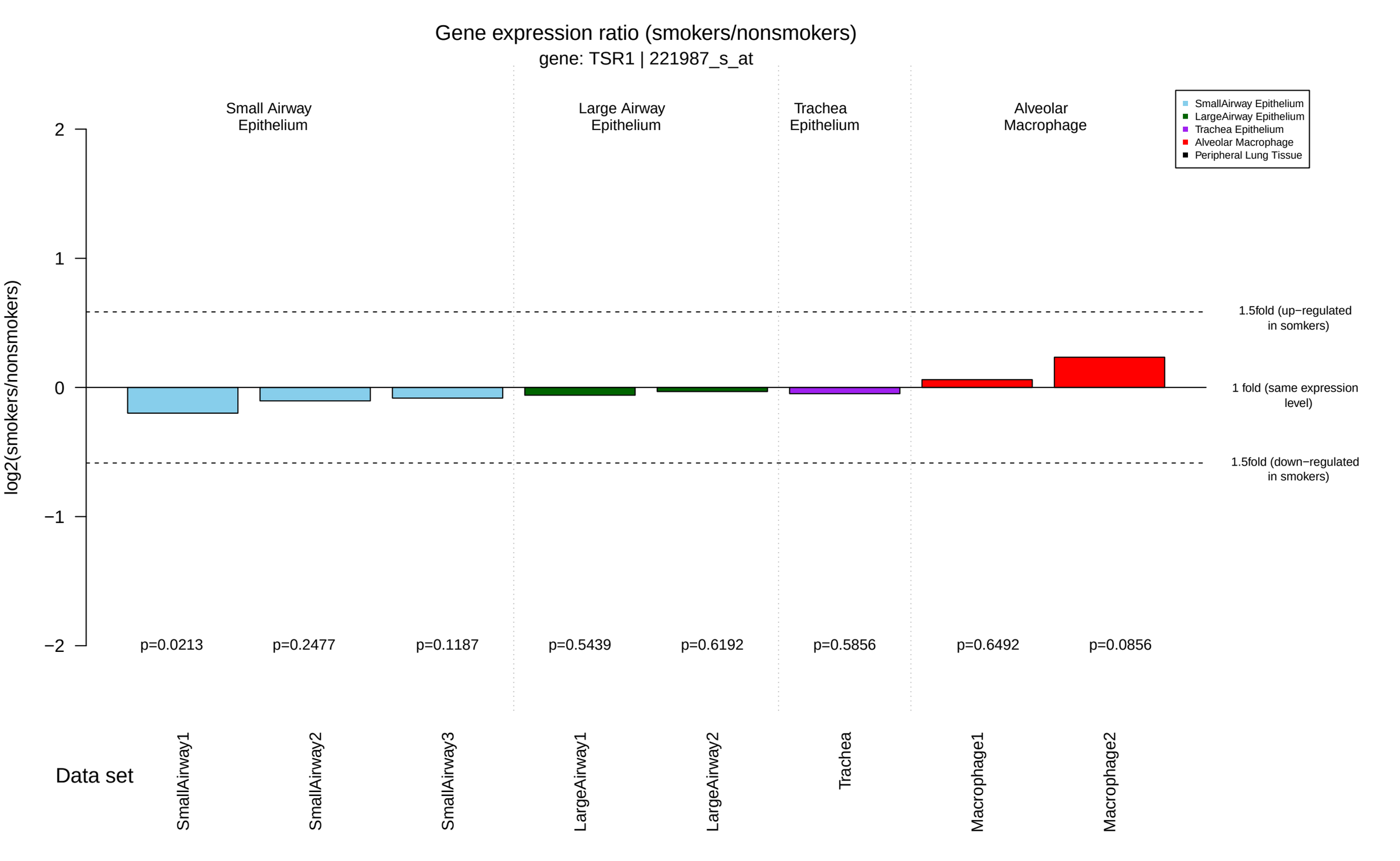
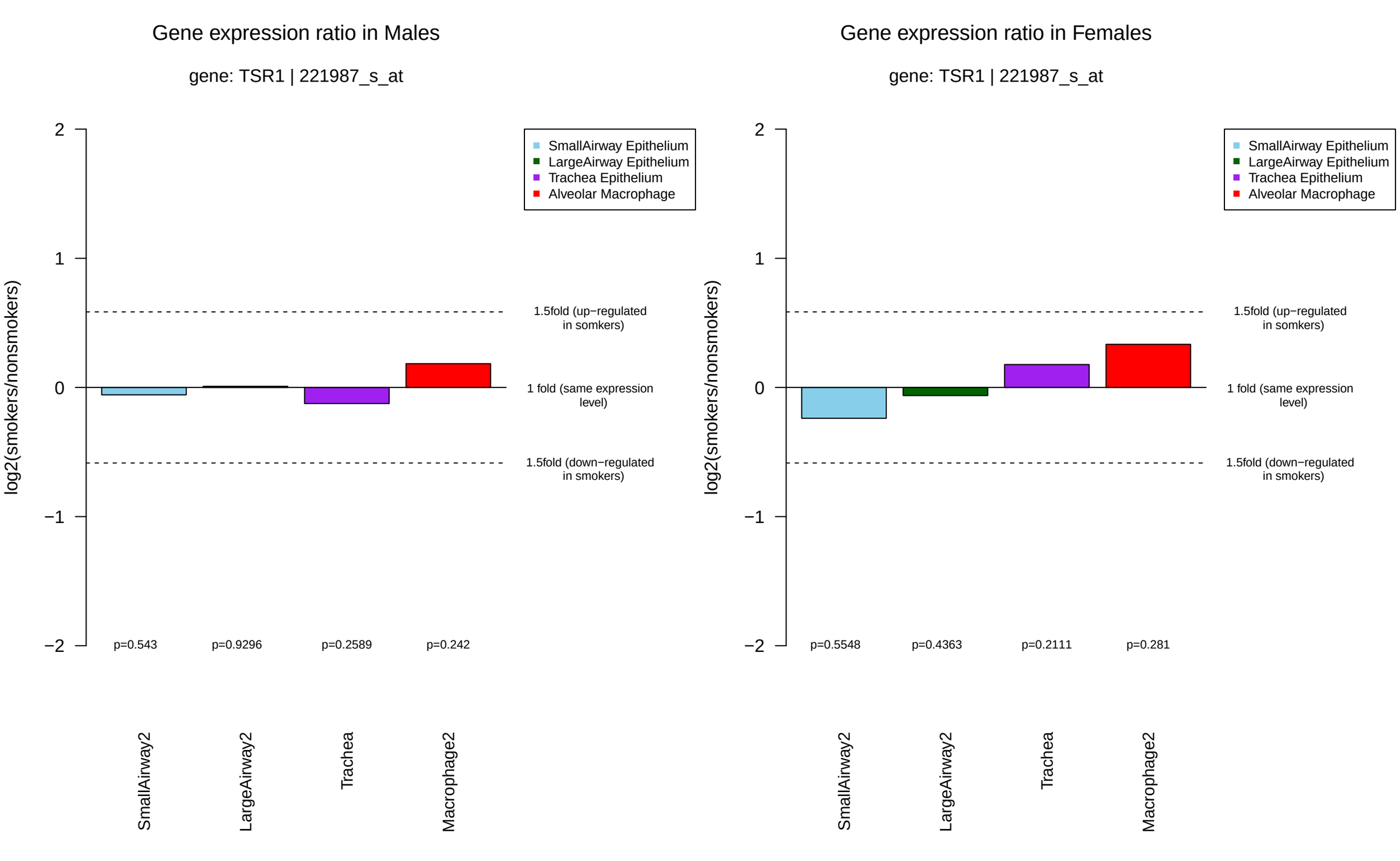
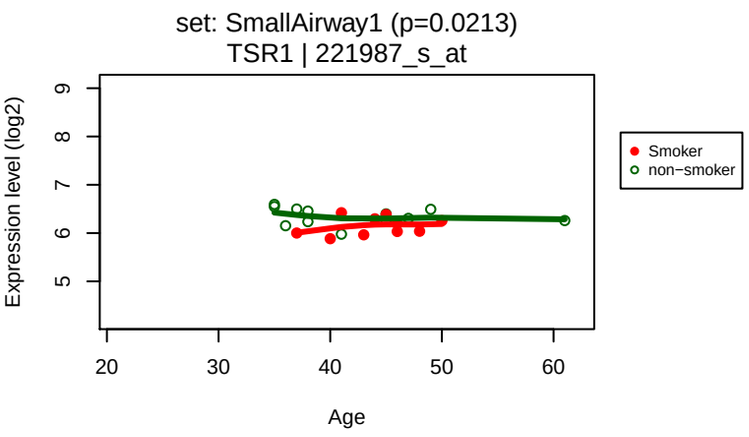
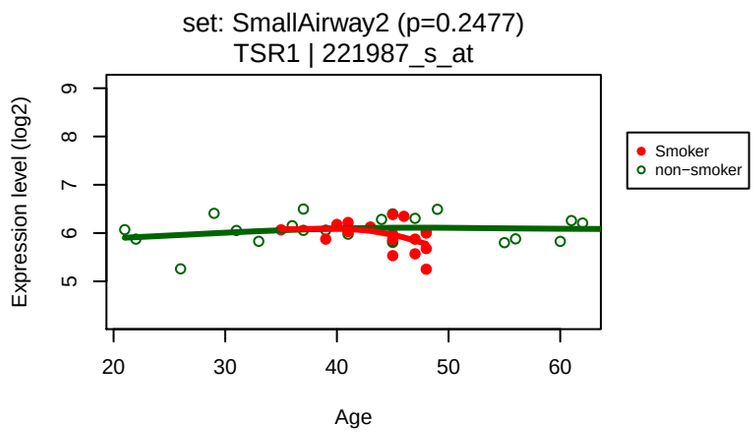
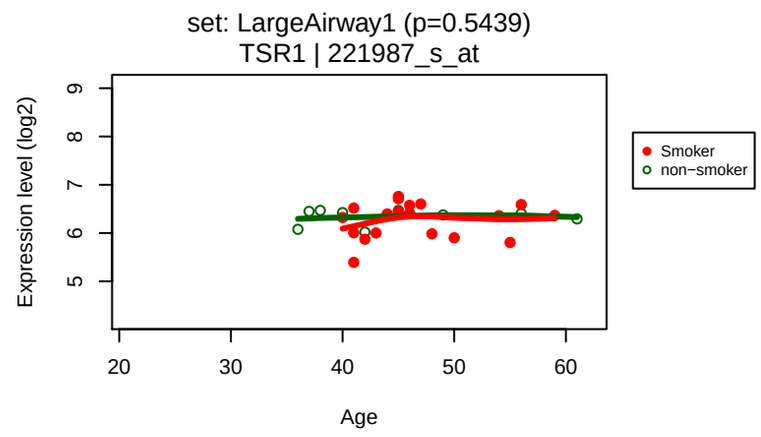

Gene expression of smokers and nonsmokers
gene: TSR1 | 221987_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0213
p=0.2477
p=0.1187
p=0.5439
p=0.6192
p=0.5856
p=0.6492
p=0.0856
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

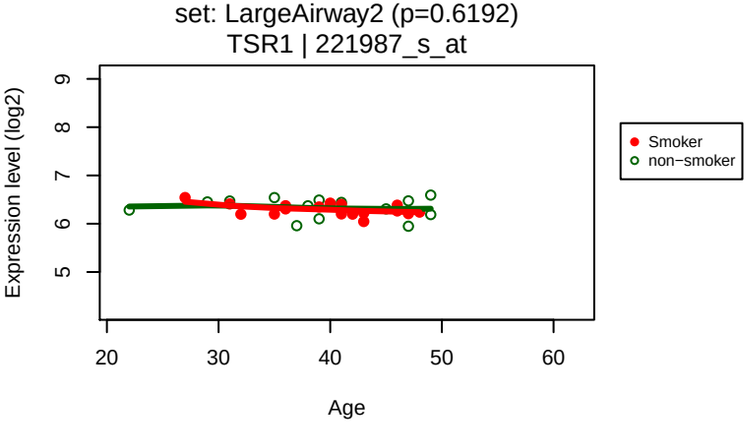

Gene expression ratio (smokers/nonsmokers)
gene: TSR1 | 221987_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0213
p=0.2477
p=0.1187
p=0.5439
p=0.6192
p=0.5856
p=0.6492
p=0.0856
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

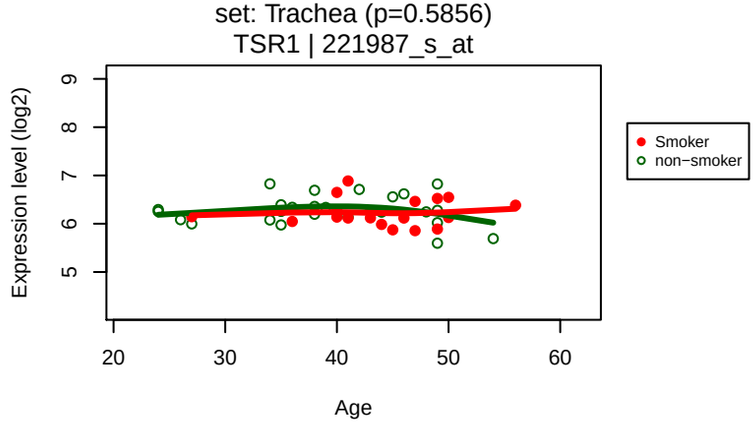

Gene expression ratio in Males
gene: TSR1 | 221987_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.543
p=0.9296
p=0.2589
p=0.242
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TSR1 | 221987_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5548
p=0.4363
p=0.2111
p=0.281
SmallAirway2
LargeAirway2
Trachea
Macrophage2

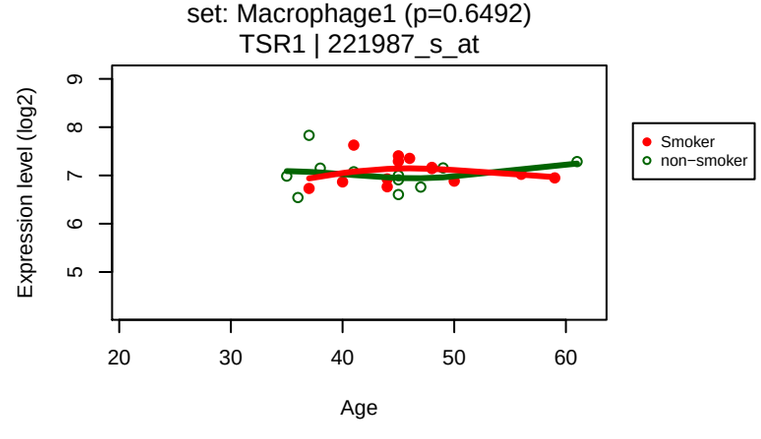

set: SmallAirway1 (p=0.0213)
TSR1 | 221987_s_at
Expression level (log2)
Age
Smoker
non−smoker

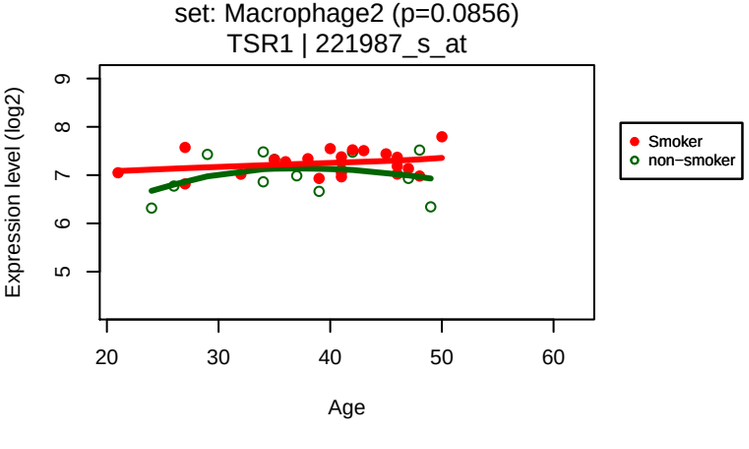

set: SmallAirway2 (p=0.2477)
TSR1 | 221987_s_at
Expression level (log2)
Age
Smoker
non−smoker

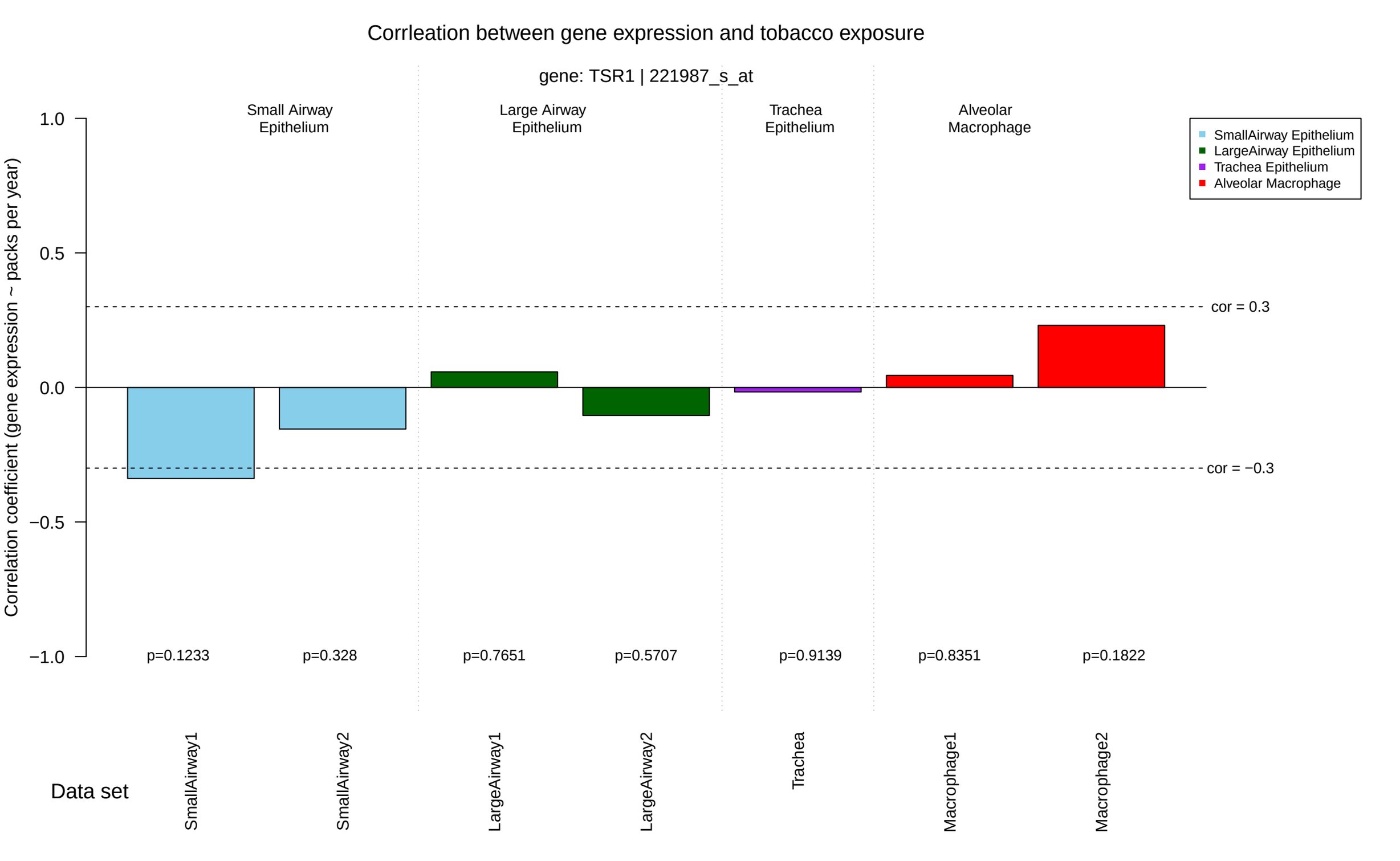

set: LargeAirway1 (p=0.5439)
TSR1 | 221987_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6192)
TSR1 | 221987_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5856)
TSR1 | 221987_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6492)
TSR1 | 221987_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0856)
TSR1 | 221987_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TSR1 | 221987_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1233
p=0.328
p=0.7651
p=0.5707
p=0.9139
p=0.8351
p=0.1822
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2